

# What we're doing

Every MarinerFold eval so far scores contact prediction. contacts-and-crops-v1 (#130) encodes real 3D positions — Pass-1 coarse 10 Å boxes plus Pass-2 fine 0.1 Å crops — so a model trained on it should be able to emit an actual structure.

This experiment asks whether it can, scored the way structure predictors are normally scored: CA-RMSD, all-atom RMSD, IDDT, IDDT-CA and TM-score against the 554-protein eval set from #74/#78.

# Two components, two gates

Component 1 — inference (document to coordinates) — is plans-first by design. Five approaches are written up in PLANS.md for discussion before anything is built.

Component 2 — scoring — is done. It takes a directory of predicted coordinate files in a documented one-PDB-per-protein contract and needs to know nothing about how they were produced. Ground truth, model predictions and baselines all speak the same contract.

# Ground truth: 554/554, and one library bug

Full-atom ground truth for all 554 records: 777,459 heavy atoms, zero failures, minimum sequence-alignment identity 0.95, lengths 30 to 761.

Building it surfaced a real bug in the shared library. The coordinate parse layer joins the pyconfind residue list to the gemmi atom walk on (chain, residue number), and the two libraries spell a blank author chain id differently — gemmi "", pyconfind "\_". Structures with no chain id joined nothing and produced a document with an empty coordinate section, silently. AFDB is always chain A so the training corpus never hit it; 19 of the 554 eval proteins did. Fixed, with the parse layer now raising instead of emitting a coordinate-free document.

# Partial predictions are the normal case

About 96% of contacts-and-crops-v1 documents are truncated by construction, so the harness reports two families of metric and keeps them apart.

Coverage-penalized (IDDT, IDDT-CA, TM-score) take their denominators from the ground truth, so an atom the model never placed costs score. These are the model-comparison numbers.

Covered-only (RMSD, and the "covered" IDDT variants) are computed over the atoms that were placed. A predictor that emits three atoms perfectly scores 0.0 A RMSD, so these are meaningless without the coverage columns beside them.

A record with no prediction file is scored as a total miss, not skipped.

# The headline so far: the format's ceiling

Before any model number means anything, we measured what a perfect model could score. Degrade the ground truth to each of the format's resolution tiers and run it through the same harness.

The 0.1 Å digit vocabulary costs nothing: IDDT 1.000, TM 1.000. But every atom at its correct 10 Å box centre scores only IDDT 0.32 / TM 0.51, and one realistic document — 65% of atoms boxed, 25% refined — tops out at IDDT 0.17 / TM 0.41.

IDDT falls as coverage squared (a contact needs both its atoms); TM-score falls linearly (a residue needs only itself).

# What that implies for inference

An inference plan that samples one document per protein is competing for IDDT 0.17. A result from such a plan cannot distinguish "the model cannot fold" from "the format did not get a chance", which is the central argument in PLANS.md for spending inference compute to raise the refined fraction.

Inference cost is not the constraint: a full 8192-token document for all 554 proteins is about three GPU-minutes on one H100.

The Pass-2 refined fraction is the entire ballgame. Going from 15% to 50% refined moves the achievable IDDT from 0.36 to 0.53 and TM from 0.58 to 0.75. That is a format finding as much as an inference one.

# Status

Harness built, tested (34 tests) and validated against a measured ceiling. Inference plans written and awaiting discussion on the issue. Model scoring is blocked on that discussion; both #137/#155 checkpoints are staged and ready.

# ceiling.png

Model-free ceiling on the 554-protein eval set: ground truth degraded to each contacts-and-crops-v1 resolution tier, scored by the same harness. Left: IDDT falls as coverage<sup>2</sup>, TM-score linearly. Right: the Pass-2 refined fraction is the whole ballgame; a single document reaches IDDT 0.17 / TM 0.41.

